

Executive Business Report

End-to-End Sales Forecasting & Demand Intelligence System

Executive Summary

The objective of this project was to develop an intelligent sales forecasting and demand analysis system for a retail business using historical sales data. Four years of sales transactions were analyzed to identify trends, seasonal patterns, customer demand, and unusual sales behavior. Three forecasting models—SARIMA, Facebook Prophet, and XGBoost—were implemented and compared. Based on the implementation, XGBoost was selected as the production model due to its strong predictive performance. The project also included anomaly detection, product demand segmentation, and an interactive Streamlit dashboard.

Key Findings

Exploratory data analysis revealed clear seasonal patterns, regional differences, and category-wise sales variation. Time series decomposition confirmed trend and seasonality. XGBoost produced accurate forecasts using lag and calendar features. Category-wise and region-wise forecasting supported better inventory planning.

Three-Month Sales Forecast

The forecasting models estimated expected sales for the next three months based on historical patterns. These forecasts can help management plan procurement, inventory allocation, warehouse capacity, and budgeting while reducing the risk of overstocking and stock shortages.

Top Three Anomalies Detected

1. High sales during festive seasons and holidays.
2. Promotional campaigns and discounts causing sudden spikes.
3. Sudden sales drops possibly due to stock shortages or supply-chain disruptions.

Product Demand Segmentation

Products were grouped into four demand segments: High Volume Stable Demand, Growing Demand, Low Volume High Volatility, and Declining Demand. These segments help businesses optimize inventory and stocking strategies.

Business Recommendations

1. Adopt XGBoost for production forecasting.
2. Maintain higher inventory for stable high-demand products.
3. Continuously monitor anomalies to identify operational issues and promotional impacts.

Project Limitation

The models rely only on historical sales data. External factors such as weather, economic conditions, competitor actions, and policy changes were not included and may influence future sales.

Conclusion

The project successfully integrates forecasting, anomaly detection, clustering, and dashboard visualization into a complete demand intelligence system that supports data-driven inventory and

supply chain decisions.